

GROUP 3 FINAL PRESENTATION

EID 101D

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PROBLEM INTRODUCTION, MOTIVATION, AND GUIDING DESIGN PRINCIPLES

30 breaks per 100 miles of pipe per year [1]

1 trillion dollars in repair and maintenance costs [2]

PROBLEM INTRODUCTION, MOTIVATION, AND GUIDING DESIGN PRINCIPLES

Guiding Design Principles

Must be able to:

- Navigate through 6 – 16 in of main line water pipes (the standard diameter)
- Travel at least 20 feet per hour (most main water pipes are a minimum of 10 feet)
- Bend at least 90 degrees
- Weigh less than 20 lbs
- Have a battery life of 2 hours
- Be waterproof and minimize use of electronics

DESIGN AND BUILD CHOICES

Main Bioinspiration:



Ribbon Worms
(*Nemertea*): Proboscis-Style
Reliner, Crawling Locomotion

First Prototype - Proboscis:



First Prototype: Working Proboscis
and Inflating Sections

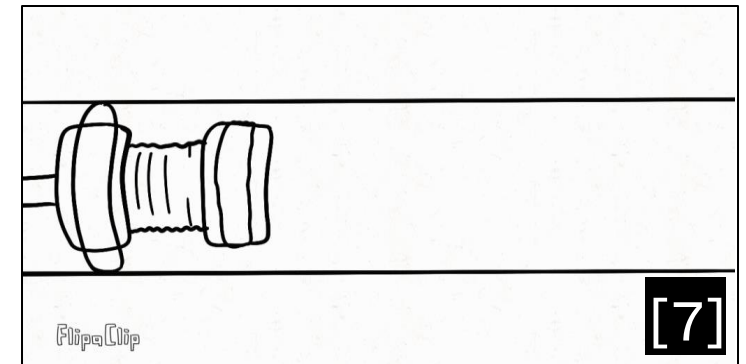


Above: Proboscis Within Pipe
Demonstration

Final Design- Locomotion:



Final Design: Working Locomotion via
Inflating Sectional Movement



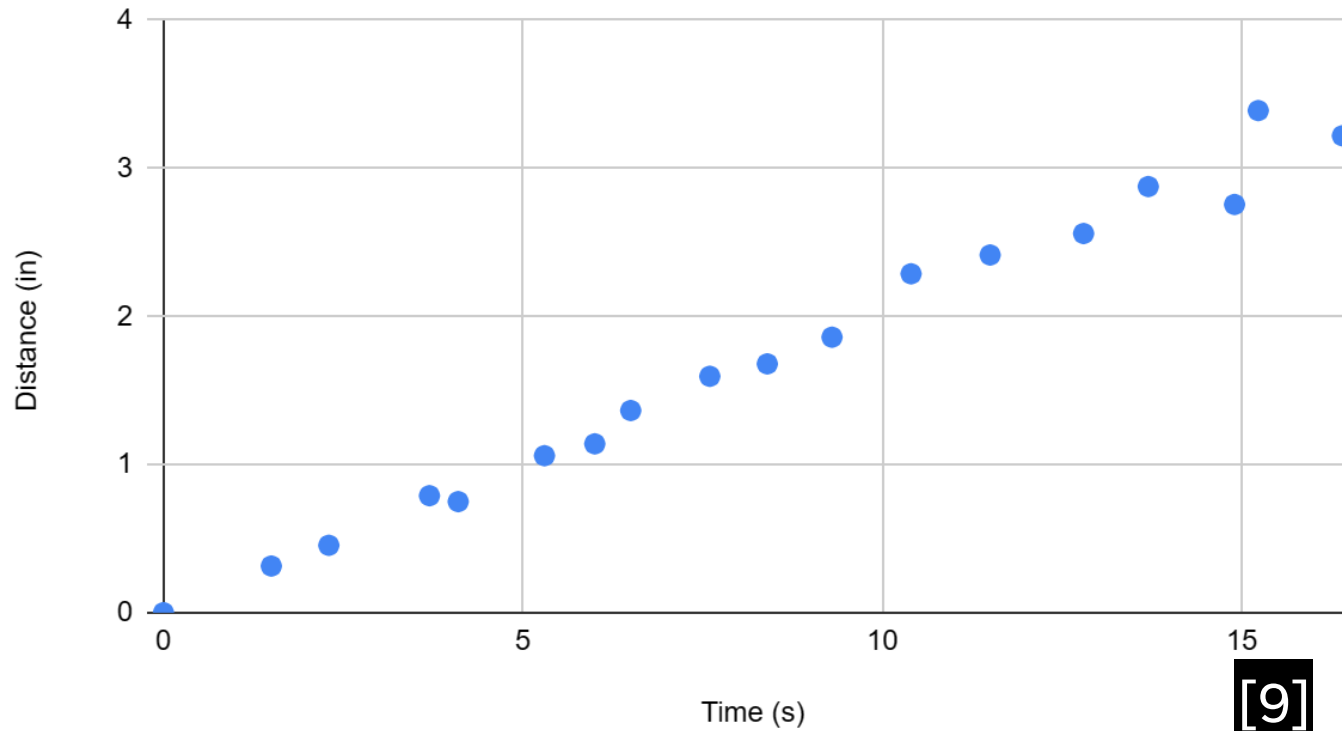
Above: Inflating Sectional
Movement Demonstration

Final Design Demonstration



RESULTS AND ANALYSIS

Distance (in) vs. Time (s)



- PSI required to pop balloon material measured to an average of **3 PSI**.
- **8 seconds** to inflate, and **8 seconds** to deflate. **48 total seconds per cycle.**
- **154 seconds per meter** (About 2.5 minutes/meter).
- About **24 meters/hour** (79 feet).
- Balloon material experienced deformation after continued use, **6.66% deformation after 20 uses**, not non-negligible amounts of change considering the consistency of the robot is important for usage.

[9]

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract shape on the left side of the slide.

THANK YOU!

WORKS CITED

- [1] Rogers, P. D., and Grigg, N. S., 2009, “Failure Assessment Modeling to Prioritize Water Pipe Renewal: Two Case Studies,” J. Infrastruct. Syst., 15(3), pp. 162–171. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2009\)15:3\(162\)](https://doi.org/10.1061/(ASCE)1076-0342(2009)15:3(162)).
- [2] Baird, G. M., and Wagner, T. B., 2020, “The High Cost of Using AWWA’s Buried No Longer Pipe Service Life Table for Capital Budgeting,” Pipelines 2020, American Society of Civil Engineers, San Antonio, Texas (Conference Held Virtually), pp. 34–44. <https://ascelibrary.org/doi/10.1061/9780784483190.005>.
- [3] TomoNews, 2015. Available: https://www.youtube.com/watch?v=_AkeEIzwjBg&ab_channel=TomoNewsUS
- [4-9] Image generated by presenters.